Neuroimmunology and Inflammation

Chapter 12

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Central question. How does the immune system influence the brain, and what role does inflammation play in trauma-related conditions?

Learning objectives

- describe the main immune signalling pathways relevant to the brain;
- explain how the brain and immune system communicate;
- evaluate evidence on inflammation in PTSD and depression;
- explain sickness behaviour and its overlap with trauma symptoms;
- describe the limits of anti-inflammatory treatment strategies.

Introduction

The immune system is not separate from the brain. It influences mood, cognition, sleep and behaviour, and the brain influences the immune system. This reciprocal relationship means that trauma-related psychological processes can leave biological traces in immune function, and that immune activation can produce psychological states that resemble trauma symptoms.

The chapter begins with basic immunology, describes neuroimmune communication, examines the evidence on inflammation in PTSD, and considers clinical implications. It is careful about causality: inflammation is unlikely to be a single cause of PTSD, but it may be a mechanism in some people and a promising target for some treatments.

> **A note on "inflammation".** *In popular usage, inflammation refers vaguely to something wrong. In immunology, it refers to a coordinated response to injury or infection, involving immune cells, cytokines and vascular changes. Chronic low-grade inflammation means persistent elevation of markers such as C-reactive protein and interleukin-6 without an obvious acute source. It is a statistical concept applied to populations, not a diagnosis for an individual.*

12.1 Innate immunity

Innate immunity is the first line of defence. It includes physical barriers, phagocytes, natural killer cells and complement proteins. It recognises molecular patterns associated with pathogens or damage through pattern-recognition receptors such as Toll-like receptors.

Innate immune activation produces cytokines—small signalling proteins including interleukins, tumour necrosis factor and interferons. These

molecules coordinate the response and can enter the brain or signal across the blood–brain barrier.

The innate immune system is fast and nonspecific. It does not require prior exposure, and it does not produce immunological memory. It is nonetheless powerful, and its activation can produce behavioural changes within hours.

12.2 Adaptive immunity

Adaptive immunity is slower and specific. It involves T and B lymphocytes, which recognise particular antigens and generate memory. It is the basis of vaccination and of long-lasting protection against pathogens.

Adaptive immunity can also produce autoimmunity, in which the response targets the body's own tissues. Some autoantibody-mediated conditions—such as anti-NMDA receptor encephalitis—present with psychiatric symptoms, which is why sudden psychiatric presentations sometimes warrant neurological investigation.

The relevance to trauma is indirect. Chronic stress can alter adaptive immune function, including antibody responses to vaccination and latency of latent viruses. These effects suggest that psychological stress has measurable immunological consequences.

12.3 Cytokines and the brain

Cytokines communicate with the brain through several routes: through circumventricular organs that lack a blood–brain barrier, through vagal signalling, through active transport, and through local production by microglia and astrocytes.

In the brain, cytokines affect neurotransmitter metabolism, synaptic plasticity and neuroendocrine function. Interleukin-1 and tumour necrosis factor can alter serotonin metabolism, affect hippocampal neurogenesis and produce behavioural changes. This is a mechanism by which systemic illness affects mood and cognition.

Peripheral cytokine changes do not map directly onto brain cytokine changes. The compartments are related but not identical, and this limits the interpretation of blood-based biomarkers.

12.4 Neuroinflammation

Neuroinflammation refers to immune activation within the central nervous system, involving microglia, astrocytes and immune cells. It is a normal response to injury or infection and becomes problematic when it is excessive or persistent.

Microglia are the resident immune cells of the brain. They survey the environment, respond to damage and remove debris. In chronic stress models, microglia show altered morphology and increased expression of inflammatory markers, and this has been linked to behavioural changes.

Human evidence on neuroinflammation is limited by the difficulty of imaging immune activity in the brain. Positron emission tomography with translocator protein ligands has been used to study microglial activation, but the method has limitations and findings are preliminary.

12.5 Sickness behaviour

Sickness behaviour is a coordinated set of responses to infection: fever, fatigue, reduced appetite, social withdrawal, anhedonia and slowed cognition. It is adaptive in the short term because it conserves energy for immune defence.

Sickness behaviour overlaps substantially with depression and with some trauma symptoms. This overlap has led to the hypothesis that some cases of depression involve inflammatory processes, and to trials of anti-inflammatory treatments.

The overlap also means that the same behaviour can have different causes. Fatigue and social withdrawal in a trauma survivor may reflect depression, inflammation, sleep disruption or all three. Assessment should not assume a single mechanism.

12.6 PTSD and inflammation

Meta-analyses report elevated levels of C-reactive protein, interleukin-6, interleukin-1 beta and tumour necrosis factor in people with PTSD compared with controls. Effect sizes are small to moderate, and there is substantial heterogeneity.

Several explanations are possible, and they are not mutually exclusive. Trauma and PTSD may cause inflammation through stress physiology. Inflammation may contribute to PTSD risk or severity. Depression, smoking, alcohol use, obesity and sleep disturbance may confound the

relationship. Childhood adversity itself is associated with elevated inflammation, which suggests a developmental pathway.

Longitudinal studies are needed to establish temporality, and they are limited. At present, inflammation is a promising marker and candidate mechanism, not a diagnostic test or proven treatment target.

A review of this literature reports elevated interleukin-6 and other inflammatory markers in PTSD, with substantial heterogeneity across studies (Hori & Kim, 2019).

12.7 Early-life adversity and immune programming

Early-life adversity is associated with altered immune function in adulthood, including elevated inflammatory markers and altered antibody responses. This may be one mechanism by which childhood adversity increases risk for a broad range of physical and mental health conditions.

Proposed mechanisms include stress-induced glucocorticoid resistance, altered HPA–immune communication, changes in the gut microbiome and accelerated biological ageing. Each has some support, none is fully established.

The implication is not that adversity determines immune fate. It is that adversity can shift developmental trajectories, and that interventions which improve social conditions may have immunological benefits.

12.8 Anti-inflammatory treatment

Anti-inflammatory drugs have been studied for depression, with modest effects in some subgroups. For PTSD, the evidence is much more limited, and no anti-inflammatory treatment is established.

Risks are significant: immunosuppression, gastrointestinal bleeding, cardiovascular events and interactions with other medication. The use of anti-inflammatory drugs for trauma-related conditions should be within research or specialist care, not general recommendation.

Non-drug approaches that reduce inflammation—exercise, sleep, diet, social connection—have better safety profiles and broader benefits. They are not a substitute for trauma treatment, but they are reasonable adjuncts.

12.9 Immune consequences of chronic stress

Chronic stress alters immune function through glucocorticoid signalling, catecholamines and the sympathetic nervous system. It can suppress some aspects of immunity, such as antiviral responses, and enhance others, such as inflammatory responses.

This dual effect helps explain why chronically stressed people may be more susceptible to infections and also more prone to inflammatory conditions. The immune system is not simply suppressed or enhanced; it is differently regulated.

Clinically, this argues for attention to infection, vaccination, wound healing and chronic disease in trauma-exposed populations, particularly where access to care is limited.

12.10 Translational caveats

The immune literature is often oversold. Headlines claim that trauma causes inflammation, that inflammation causes depression, or that a drug will cure both. The reality is more cautious.

Cytokines are pleiotropic, meaning they have many effects depending on context, concentration and target. Blood markers do not reliably reflect brain processes. Animal models do not fully capture human trauma. And the populations studied often differ in ways that affect immune measures.

A responsible account presents the evidence as suggestive rather than definitive and emphasises the need for well-powered longitudinal and intervention studies.

12.11 The gut–brain axis

The gastrointestinal tract contains a large and diverse microbial community, and it is innervated by the vagus and by enteric neurons (Cryan et al., 2019). Signals travel in both directions: the brain influences motility, secretion and permeability, and the gut influences immune function, satiety and behaviour.

Several mechanisms are plausible. Microbial metabolites, including short-chain fatty acids produced by fermentation of dietary fibre, influence epithelial integrity and immune signalling. The vagus carries information from the gut to the brainstem, where it can influence autonomic and limbic activity. Intestinal permeability, sometimes

described loosely as a "leaky gut", allows bacterial products to enter the circulation and stimulate inflammatory responses.

The evidence in humans is mostly correlational. Studies report differences in microbial composition between groups defined by trauma history, depression or inflammation, but the effect sizes are modest, the confounders are numerous (diet, medication, housing, smoking, sleep), and the direction of causality is unresolved. Probiotic trials in psychiatry have produced small and inconsistent effects.

Commercial microbiome testing and probiotic marketing frequently outrun the evidence. A clinician should be cautious about attributing trauma-related symptoms to microbial imbalance or recommending expensive supplements on that basis. General measures that support metabolic and immune health—adequate diet, sleep, exercise and treatment of gastrointestinal disease—are reasonable and do not depend on speculative mechanisms.

12.12 Measuring immune markers in trauma research

The most commonly reported markers are C-reactive protein, interleukin-6 and tumour necrosis factor. Each behaves differently, and each is affected by factors unrelated to trauma.

C-reactive protein is produced by the liver in response to interleukin-6. It rises acutely with infection or injury and is influenced by body mass, smoking, medication and chronic disease. A single value cannot distinguish trauma-related inflammation from a mild respiratory infection or metabolic condition.

Interleukin-6 is secreted in pulses, has a short half-life, and varies with time of day, sleep and exercise. A single plasma measurement is therefore a poor summary of chronic inflammatory state. Tumour necrosis factor is similar: concentrations are low, assays vary, and the signal is easily obscured by pre-analytical handling.

Measurement quality matters as much as study design. Sample timing, fasting status, anticoagulant, centrifugation delay, storage temperature, freeze–thaw cycles and assay sensitivity all influence results. Studies that do not report these details are difficult to interpret, and comparisons across laboratories require standardisation.

For research, preregistration of the primary marker, adjustment for the main confounders, reporting of assay limits and avoidance of multiple

uncorrected comparisons are minimum standards. For clinical practice, immune markers should be ordered to answer a specific medical question, not to confirm a psychological formulation.

Synthesis

Table 12.1 summarises immune pathways and trauma relevance.

Table 12.1. Immune pathways and trauma relevance

Pathway	Key mediators	Function	Trauma relevance	Evidence
Innate immunity	Cytokines, phagocytes	Rapid defence	Elevated CRP, IL-6 in PTSD	Moderate cross-sectional
Adaptive immunity	T and B cells	Specific defence, memory	Altered antibody response to stress	Limited
Microglia	Brain immune cells	Surveillance, pruning	Stress-related activation	Mostly animal
HPA–immune	Glucocorticoids	Immune modulation	Glucocorticoid resistance	Moderate
Sickness behaviour	Cytokines	Energy conservation	Overlap with depression	Conceptual

Composite case study

A patient with PTSD and depression has elevated inflammatory markers and asks whether she has "brain inflammation". The clinician explains that elevated markers are common in many conditions, are not specific to brain inflammation, and do not currently guide treatment. The plan addresses sleep, exercise, smoking and depression, with trauma-focused therapy as the core treatment.

Review and discussion questions

How do cytokines communicate with the brain?

Why is elevated CRP not a diagnostic test for PTSD?

What is sickness behaviour, and how does it overlap with depression?

Why are longitudinal studies needed in psychoneuroimmunology?

What are the risks of anti-inflammatory treatment for trauma-related conditions?

How might early adversity alter immune development?

Research assignments

- Design a longitudinal study of inflammation and PTSD that controls for smoking and sleep.
- Critically appraise one trial of an anti-inflammatory drug in depression.
- Write a plain-language explanation of the immune findings for clinicians without immunology training.

Chapter summary

The immune system communicates with the brain through cytokines, vagal pathways and microglial activity, influencing mood, cognition and behaviour. PTSD is associated with small elevations in inflammatory markers, but causality and clinical utility are unresolved. Sickness behaviour overlaps with depression and trauma symptoms. Anti-inflammatory treatment is not established for PTSD, and non-drug approaches with broader benefits should be considered. Claims in this field should be proportionate to the evidence.

References

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